D1

Find the worst-case DC output voltages of the inverting amplifier shown in Figure 2.34a for $v_{in}$ = 0. The maximum bias current is 200 nA, the maximum offset current magnitude is 50 nA, and the maximum offset voltage magnitude is 4mV.

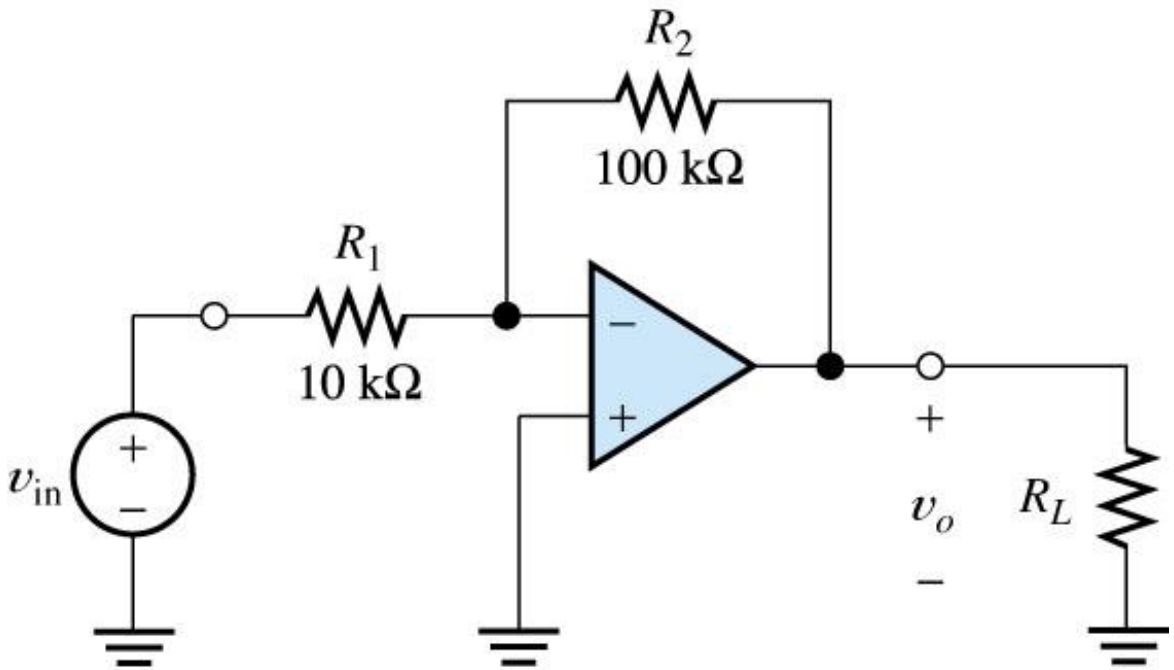



Figure 2.34a

D2

Sometimes an AC-coupled amplifier is needed. The circuit shown in Figure P2.59 is a poor way to accomplish AC coupling. Explain why.
(*Hint: Consider the effect of bias current*.)
Show how to add a component (including its value) so that bias current has no effect on the output voltage of this circuit.

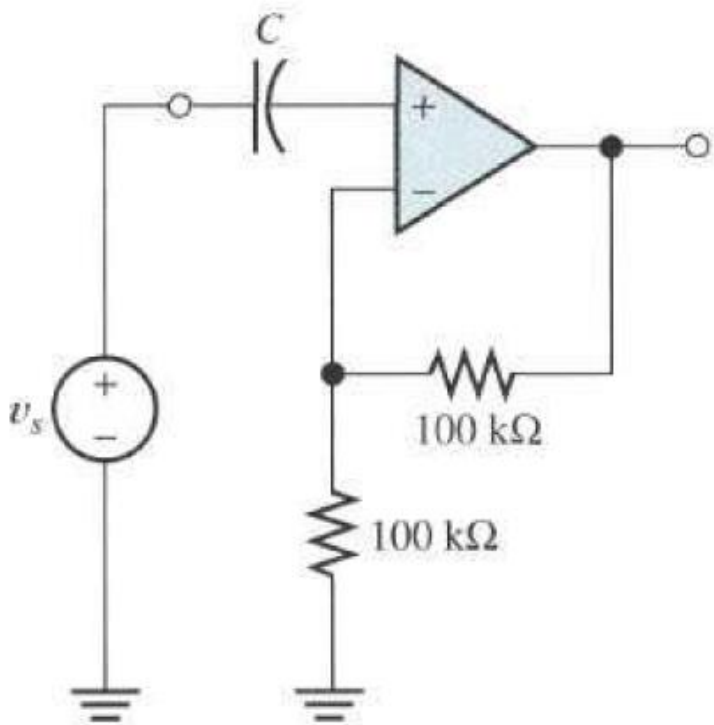
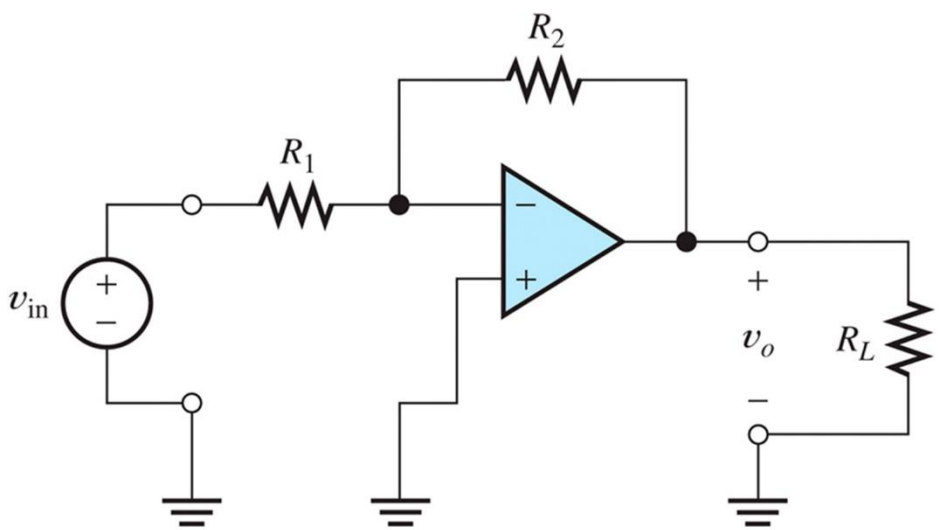



Figure P2.59

D3

Consider the amplifier shown in the schematic below. With zero DC input voltage from the signal source, it is desired that the DC output voltage be no greater than 100 mV in magnitude.

a) Ignoring other DC imperfections, what is the maximum offset voltage allowed for the op amp?
b) Ignoring other DC imperfections, what is the maximum bias current allowed for the op amp?
c) Show how to add a resistor to the circuit, including its value, so that the effects of the bias currents cancel.
d) Assuming that the resistor of part c) is in place, and ignoring offset voltage, what is the maximum offset current allowed for the Op Amp?




$R_1$ = 10 kΩ

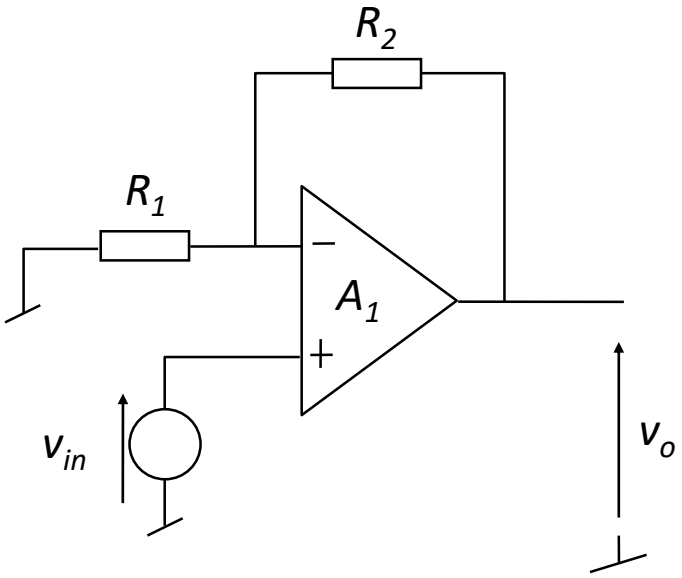
$R_2$ = 100 kΩ

$v_{in} = V_{in} \cdot \sin(\omega t)$

D4

En ikke-inverterende forstærker er opbygget med en μA741C som Op Amp $A_1$. Kredsløbet befinder sig ved 25 °C og er forsynet med ±15 V. Der kan ses bort fra evt. generatormodstand for $v_{in}$. $A_1$ kan regnes ideel bortset fra offset- og biasfejl.

Beregn den maksimale fejl der kan forekomme på udgangen hidrørende fra offset- og biasfejl.




$R_1$ = 10 kΩ

$R_2$ = 470 kΩ

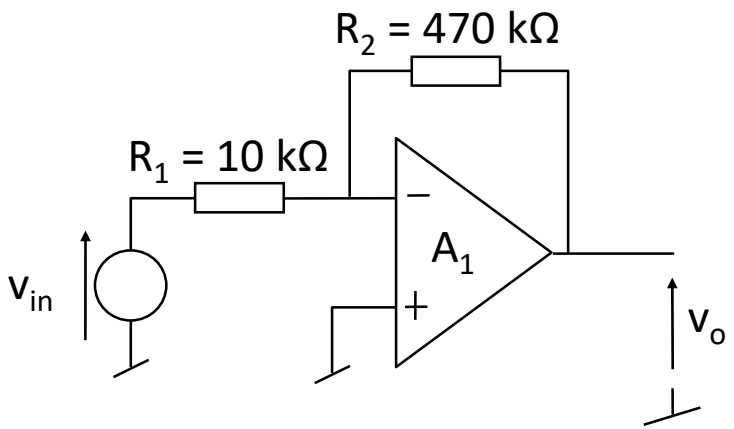
D5

En inverterende forstærker er opbygget med en μA741C som Op Amp $A_1$. Kredsløbet befinder sig ved 25 °C og er forsynet med ±15 V. Der kan ses bort fra evt. generatormodstand for $v_{in}$. $A_1$ kan regnes ideel bortset fra offset- og biasfejl.

1) Beregn den maksimale fejl der kan forekomme på udgangen hidrørende fra offset- og biasfejl.

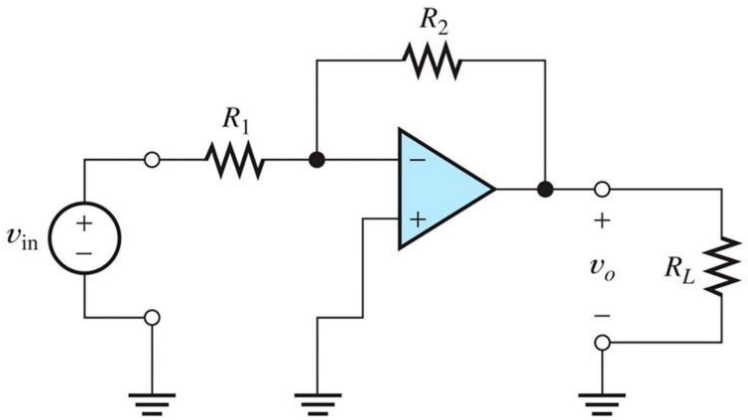
For at mindske fejlen anbringes en modstand $R_3$ i den ikke-inverterende terminal.

2) Hvad er ideen bag dette princip – hvordan kan $R_3$ bidrage til at mindske fejlen?
3) Hvilken værdi skal $R_3$ have, såfremt fejlen på udgangen skal mindskes mest muligt?
4) Hvad bliver nu den maksimale fejl på udgangen?



D6

En operationsforstærker er koblet som inverterende forstærker:




$R_1 = 10\ k\Omega$

$R_2 = 20\ k\Omega$

$R_L = 10\ k\Omega$

Operationsforstærkeren kan regnes ideel på nær en 1. ordens frekvensrespons for åbensløjfeforstærkningen:

Åbensløjfe DC forstærkning $A_{0OL}$ = 100 dB
Dominerende pol i åbensløjfeforstærkning $f_{BOL}$ = 10 Hz

*(Opgave D6 fortsætter på næste side)*

Opgaven går i det følgende ud på at øve det logaritmiske Bode Plot samt forstå hvordan dette kan bruges til at overskue frekvensforholdene for en forstærker.

1) Bestem bogstavudtryk og talværdier for α, β og $\frac{\alpha}{\beta}$.

Lukketsløjfeforstærkningen kan skrives på følgende facon:

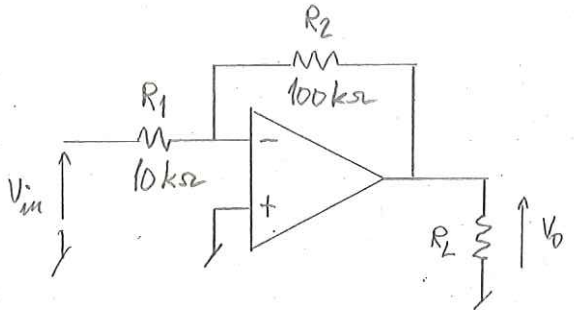
$$A_{CL} = \frac{v_o}{v_{in}} = \frac{\alpha}{\beta} \cdot Kf$$

2) Hvordan ser denne sammenhæng ud på logaritmisk form - altså $20\log|A_{CL}|$?
3) Vurdér fejlfaktoren $K_f$ for $\beta A_{OL} >> 1$ og for $\beta A_{OL} << 1$.
4) Skitsér Bode Plot (kun amplitudekarakteristik) for følgende størrelser:

| | | | | | | |
|---|---|---|---|---|---|---|
| $A_{OL}$ | $\alpha$ | $\frac{1}{\beta}$ | $\frac{\alpha}{\beta}$ | $\beta A_{OL}$ | $K_f$ | $A_{CL}$ |

*Det kan anbefales at gøre det i den nævnte rækkefølge* ☺

D1

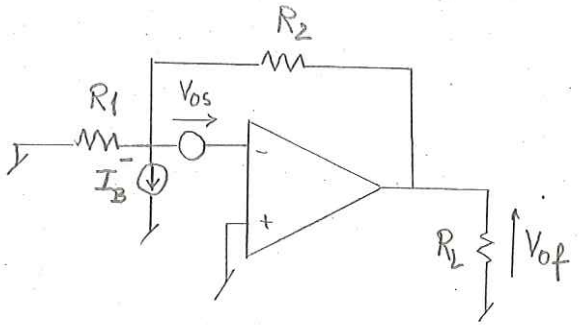



$I_{B_{max}} = 200\ nA$

$I_{OS_{max}} = \pm 50\ nA$

$V_{OS_{max}} = \pm 4\ mV$

DC-akv. med $V_{in} = 0$:




$I_B^- = I_B - \frac{I_{OS}}{2}$

$e^+ = 0$ (ingen modstand fra den ikke-inverterende indg. til stel)

$e^- = V_{OS} - I_B^-(R_1 \| R_2) + V_{of}\frac{R_1}{R_1+R_2}$ (superposition)

$e^+ = e^-$

$\Downarrow$

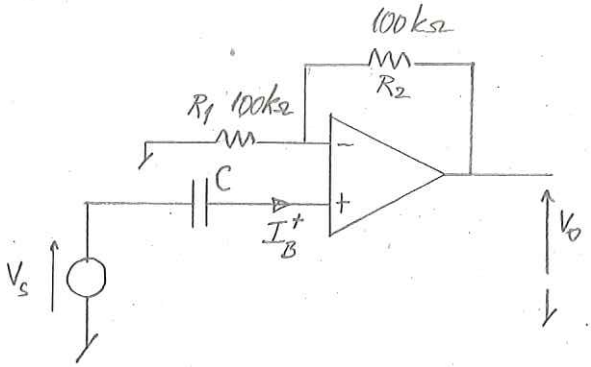
$$V_{of} = \frac{R_1+R_2}{R_1}\left(I_B^-(R_1\|R_2) - V_{OS}\right) = R_2 \cdot I_B^- - \frac{R_1+R_2}{R_1}\cdot V_{OS}$$

Ved indsættelse findes max fejl:

$$V_{of_{max}} = R_2 \cdot \underset{\text{(Singular)}}{I_{B_{max}}^-} - \frac{R_1+R_2}{R_1}\cdot \underset{\text{(Mest negativ)}}{V_{OS_{min}}}$$

$$= 10^5 \cdot \left(200 - \frac{(-50)}{2}\right)\cdot 10^{-9} - \frac{10+100}{10}\cdot(-4\cdot 10^{-3})$$

$$= 22{,}5\cdot 10^{-3} + 44{,}0\cdot 10^{-3} \approx \underline{\underline{66{,}5\ mV}}$$




Op. Amp. tvinges til at trække sin biasstrøm $I_B^+$ gennem en DC-mæssig uendelig høj impedans:

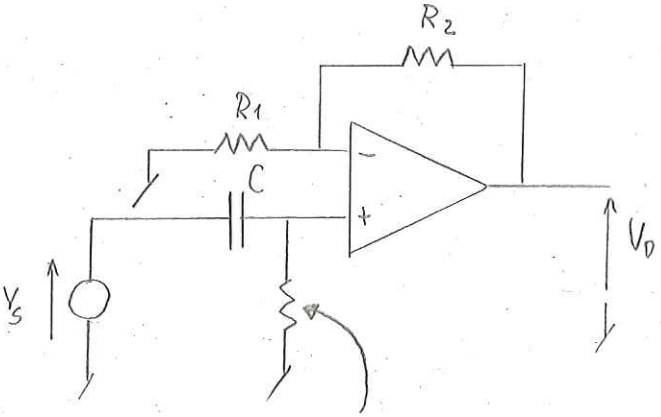
$$\left|\frac{1}{j\omega C}\right|_{\omega=0} \longrightarrow \infty \Rightarrow e^+ \longrightarrow \infty$$

og Op. Amp. vil gå i mætning.

Løsningen er at give en returvej for $I_B^+$, og såfremt der samtidig skal

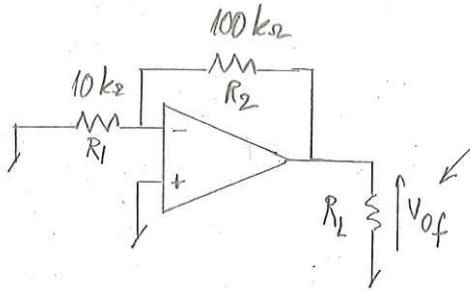
Biaskompenseres

fås følgende løsning:




$$R_B = R_1 // R_2 = 100 // 100 = \underline{\underline{50k\Omega}}$$

## D3 Figur P2.54 med $v_s(t) = 0$

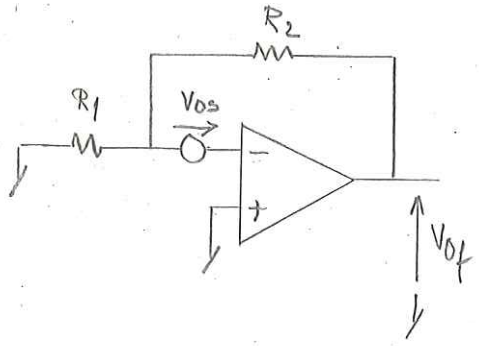



Fejlspænding indtr. fra Offset- og biasfejl.

$$\beta = \frac{R_1}{R_1 + R_2} = \frac{10}{10+100} = \frac{1}{11} \approx 0{,}0909$$

I hele opgaven gælder: $V_{of\,max} = 100\ mV$

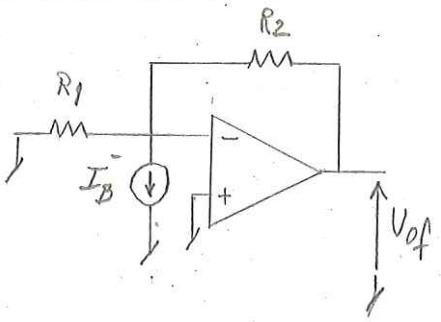
a) $I_B = 0;\ I_{OS} = 0$




$$V_{of} = -\frac{1}{\beta} \cdot V_{OS} \Rightarrow$$

$$|V_{OS}|_{max} = \beta \cdot V_{of\,max} = \frac{100 \cdot 10^{-3}}{11} = \underline{\underline{9{,}09\ mV}}$$

b) $V_{OS} = 0;\ I_{OS} = 0$




$$I_{OS} = 0 \Rightarrow I_B^- = I_B \Rightarrow$$

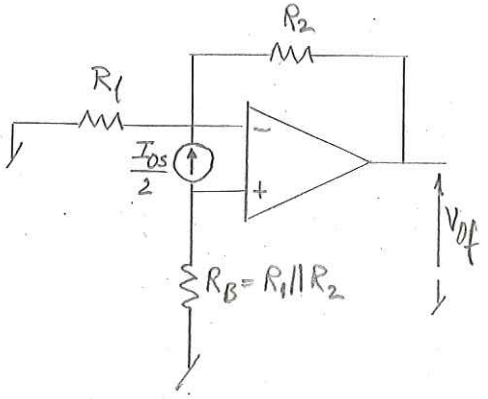
Pga. negative feedback

$$e^- = I_B (R_1 \| R_2) + V_{of} \cdot \frac{R_1}{R_1 + R_2} = e^+ = 0 \Rightarrow$$

$$V_{of} = R_2 \cdot I_B \Rightarrow I_{B\,max} = \frac{V_{of\,max}}{R_2} = \frac{100 \cdot 10^{-3}}{10^5} = \underline{\underline{1\ \mu A}}$$

c) Biaskompensering med $R_B = R_1 \| R_2 = 10 \| 100 = 9{,}09\ k\Omega$ placeret i den ikke-inverterende indgang (se evt. næste sp.).

d) $V_{OS} = 0$; Biaskompensering




$$e^- = \frac{I_{OS}}{2}(R_1 \| R_2) + V_{of} \frac{R_1}{R_1 + R_2}$$

$$e^+ = -\frac{I_{OS}}{2} \cdot R_B = -\frac{I_{OS}}{2} \cdot (R_1 \| R_2)$$

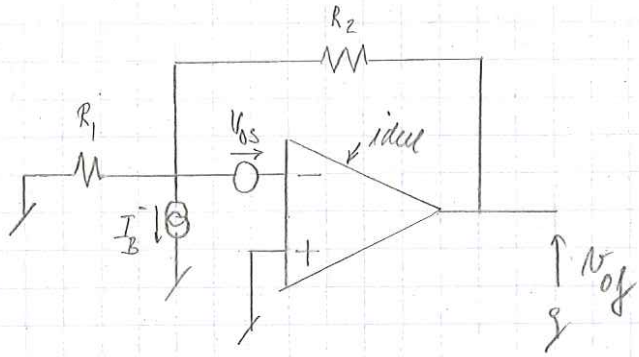
$$e^+ = e^- \Rightarrow$$

$$V_{of} = \frac{R_1 + R_2}{R_1}\left(-\frac{I_{OS}}{2}(R_1 \| R_2) - \frac{I_{OS}}{2}(R_1 \| R_2)\right) = -I_{OS} \cdot R_2 \Rightarrow$$

$$|I_{OS}|_{max} = \frac{V_{of\,max}}{R_2} = \frac{100 \cdot 10^{-3}}{10^5} \approx \underline{\underline{1\ \mu A}}$$

D4

Akv.




$I_B^+ = I_B + \frac{I_{OS}}{2}$

$I_B^- = I_B - \frac{I_{OS}}{2}$

$R_1 = 10\,k\Omega$

$R_2 = 470\,k\Omega$

$e^+ = 0$

$e^- = V_{OS} + \frac{R_1}{R_1+R_2} \cdot v_{of} - I_B^-(R_1 \| R_2)$

$e^- = e^+ \Rightarrow \quad v_{of} = \frac{R_1+R_2}{R_1}\left[I_B^- \cdot (R_1 \| R_2) - V_{OS}\right]$

$= \underbrace{I_B^-}_{\text{Singular}} \cdot R_2 - \frac{R_1+R_2}{R_1} \cdot \underbrace{V_{OS}}_{\text{Dual}}$

$v_{of} = v_{of\,max}$ for:

$I_B^- = I_{B\,max}^-$

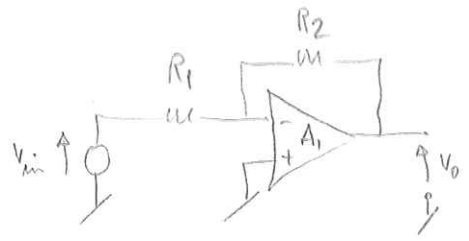
$V_{OS} = V_{OS\,min}$ („mest negativ")

$$\left.\begin{aligned} I_{B\,max}^- &= 500 - \left(-\frac{200}{2}\right) = 600\ nA \\ V_{OS\,min} &= -6\,mV \end{aligned}\right\} \Rightarrow$$

$v_{of\,max} = 600 \cdot 10^{-9} \cdot 470 \cdot 10^3 - \frac{10+470}{10} \cdot (-6 \cdot 10^{-3})$

$= 282{,}0 \cdot 10^{-3} + 288{,}0 \cdot 10^{-3} = \underline{\underline{570\,mV}}$

D5



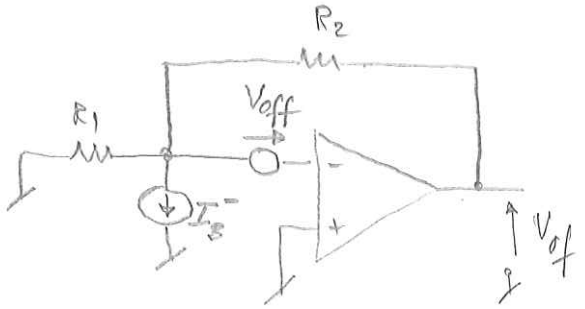

T = 25 °C

$A_1$: µA741C

$V^+ = -V = 15V$

Signalsp.

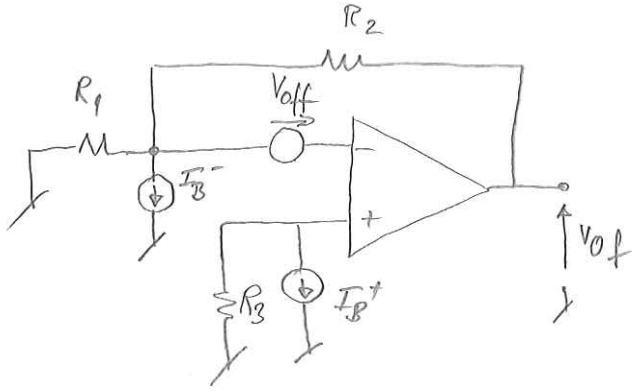
Sp.1 Ækv. diagram til beregning af fejlspænding $V_{of}$ ($V_{in} = 0$):




Identisk med ækvivalent-diagram i Opgave 4

$$V_{of\,max} = \underline{\underline{570\,mV}}$$

Sp.2 Ækv. diagram med $R_3$ indsat:




$$I_B^+ = I_B + \frac{I_{off}}{2}$$

$$I_B^- = I_B - \frac{I_{off}}{2}$$

$$e^+ = -I_B^+ \cdot R_3$$

$$e^- = V_{off} - I_B^-(R_1 \| R_2) + V_{of} \cdot \frac{R_1}{R_1 + R_2}$$

$$e^- = e^+ \Rightarrow V_{of} = \frac{R_1+R_2}{R_1}\left[I_B^-(R_1\|R_2) - I_B^+ \cdot R_3 - V_{off}\right]$$

$$= \frac{R_1+R_2}{R_1}\left[\left(I_B - \frac{I_{off}}{2}\right)(R_1\|R_2) - \left(I_B + \frac{I_{off}}{2}\right)\cdot R_3 - V_{off}\right]$$

$$= \frac{R_1+R_2}{R_1}\left[I_B(R_1\|R_2 - R_3) - \frac{I_{off}}{2}(R_1\|R_2 + R_3) - V_{off}\right]$$

Dette led kan udelades

Sp.3 Vælges $R_1 \| R_2 = R_3$ forsvinder det første led med biasstrømmen $I_B$: BIASKOMPENSERING

$$V_{of}\Big|_{R_1 \| R_2 = R_3} = \frac{R_1+R_2}{R_1}\left[-\frac{I_{off}}{2}(R_1 \| R_2 + R_1 \| R_2) - V_{off}\right]$$

$$= -I_{off}\cdot R_2 - \frac{R_1+R_2}{R_1}\cdot V_{off}$$

Generelt er $|I_B| \ll |I_{off}|$

Bemærk at nu er hele fejlen dual.

Sp.4 Beregning af max fejl:

$\mu A741C$: $I_{off\,max} = \pm 200\,nA$, $V_{off\,max} = \pm 6\,mV$ $\Rightarrow$

$$V_{of\,max} = -(-200\cdot 10^{-9})\cdot 470\cdot 10^{3} - \frac{10+470}{10}\cdot(-6\cdot 10^{-3}$$

$$= 94\cdot 10^{-3} + 288\cdot 10^{-3} = \underline{\underline{382\ mV}}$$

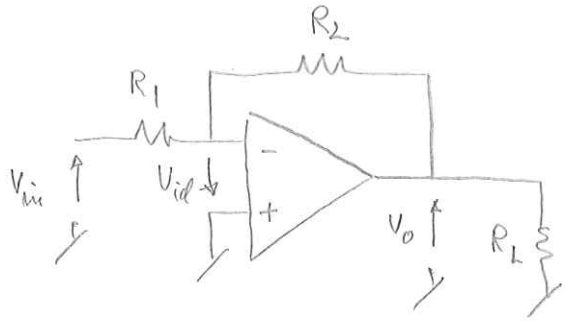
$V_{off}$ giver her det største bidrag pga. en "høj" forstærkning.

---

Princip i biaskompensering:

Den inverterende og den ikke-inverterende terminal skal "se ud i samme DC-modstand".

D6




$R_1 = 10\,k\Omega$

$R_2 = 20\,k\Omega$

$R_L = 10\,k\Omega$

$$A_{OL}(f) = \frac{A_{0OL}}{1 + j\frac{f}{f_{BOL}}} \quad ; \quad A_{0OL} = 100\,dB = 10^5 \quad ; \quad f_{BOL} = 10\,Hz$$

Sp. 1

$$\alpha = \left.\frac{V_{id}}{V_{in}}\right|_{V_0=0} = \underline{\underline{\frac{-R_2}{R_1+R_2}}} = \frac{-20}{10+20} \cong \underline{\underline{-0{,}667}}$$

$$\beta = \left.\frac{-V_{id}}{V_0}\right|_{V_{in}=0} = \underline{\underline{\frac{R_1}{R_1+R_2}}} = \frac{10}{10+20} \cong \underline{\underline{0{,}333}} \Rightarrow \frac{1}{\beta} = 3$$

$$\frac{\alpha}{\beta} = -\frac{R_2}{R_1+R_2}\cdot\frac{R_1+R_2}{R_1} = \underline{\underline{-\frac{R_2}{R_1}}} = -\frac{20}{10} = \underline{\underline{-2}}$$

Sp 2

$$A_{CL} = \frac{V_0}{V_{in}} = \frac{\alpha}{\beta}\cdot K_f \Rightarrow 20\log|A_{CL}| = 20\log|\alpha| + 20\log\left|\frac{1}{\beta}\right| + 20\log|K_f|$$

$$\alpha = -0{,}667 \Rightarrow 20\log|\alpha| \cong -3{,}52\,dB$$

$$\frac{1}{\beta} = 3 \Rightarrow 20\log\left|\frac{1}{\beta}\right| \cong 9{,}54\,dB$$

$$\frac{\alpha}{\beta} = -2 \Rightarrow 20\log\left|\frac{\alpha}{\beta}\right| = 20\log|\alpha| + 20\log\left|\frac{1}{\beta}\right| \cong 6{,}02\,dB$$

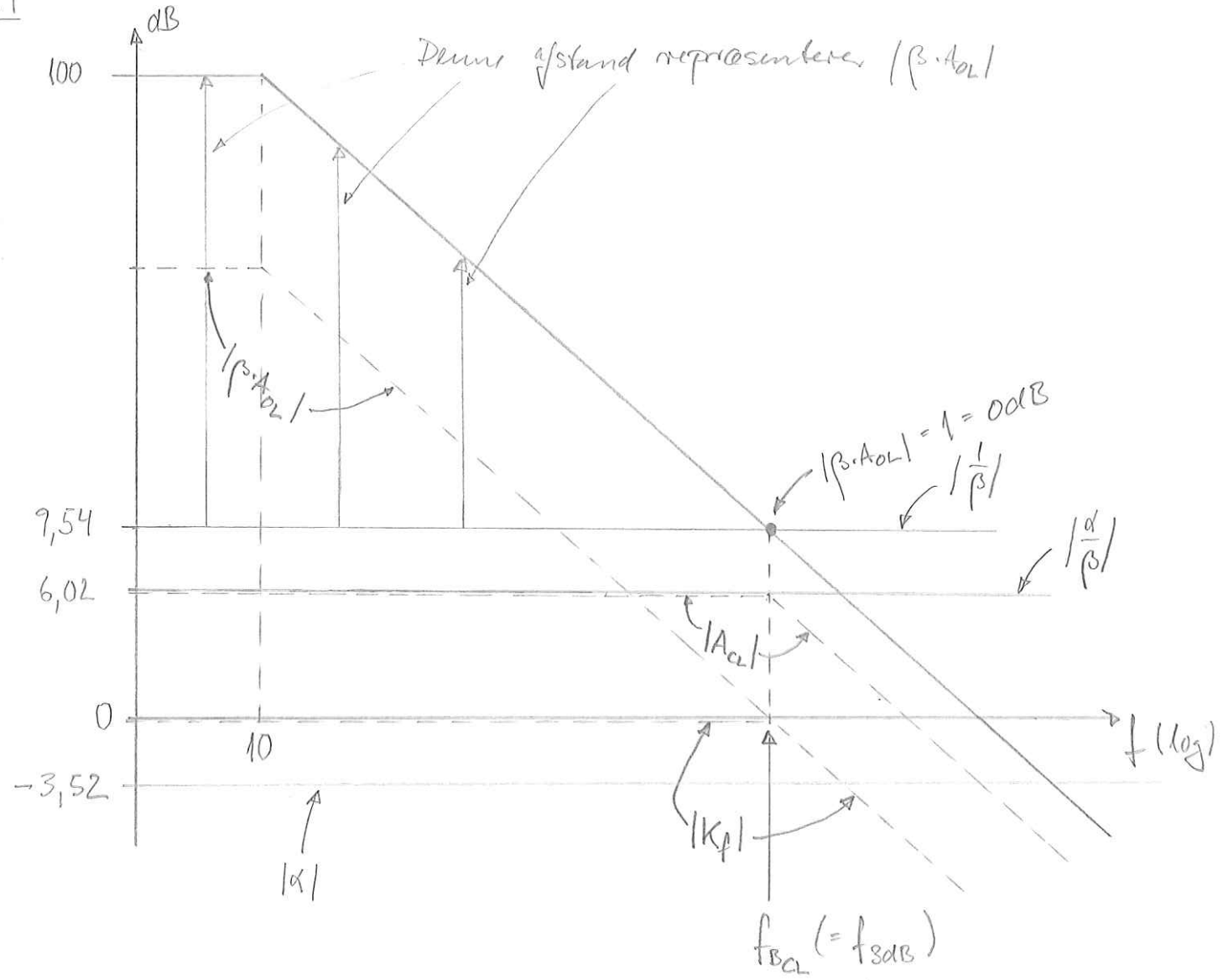
Sp. 3

$$K_f = \frac{1}{1 + \frac{1}{\beta \cdot A_{OL}}}$$

$$\beta \cdot A_{OL} \gg 1 \quad \Rightarrow K_f \cong 1 = 0\,dB$$

$$\beta \cdot A_{OL} \ll 1 \quad \Rightarrow K_f \cong \beta \cdot A_{OL}$$

Sp. 4




$|A_{CL}|$ fås som (den grafiske) addition $|\frac{\alpha}{\beta}| + |K_f|$